

Table 1: Quantum-resource comparison for a single in-place point addition on binary elliptic curves

n	Source	Strategy	#Toffoli	#CNOT	Width	Toffoli depth	NCT depth	DW-cost	TDW-cost
163	[JSB+25] ours	FLT-in	193354	2417882	53134	46	5888	1.17×2^{28}	2444164
		AC-mul + JSB25-inv	40190	7375426	11362	46	32656	1.38×2^{28}	522652
		AC-mul + AC-inv	40190	7065450	13174	42	29872	1.47×2^{28}	553308
233	[JSB+25] ours	FLT-in	303970	4326952	82899	50	7186	1.11×2^{29}	4144950
		AC-mul + JSB25-inv	64834	16961230	18133	50	47458	1.60×2^{29}	906650
		AC-mul + AC-inv	64834	16440286	20815	42	40098	1.55×2^{29}	874230
283	[JSB+25] ours	FLT-in	534762	7563802	144672	54	11149	1.50×2^{30}	7812288
		AC-mul + JSB25-inv	87302	27842954	24202	54	61535	1.39×2^{30}	1306908
		AC-mul + AC-inv	87302	26961190	27538	46	53571	1.37×2^{30}	1266748
571	[JSB+25] ours	FLT-in	1871402	29759716	500450	62	23766	1.38×2^{33}	31027900
		AC-mul + JSB25-inv	215282	142024994	58818	62	139814	1.91×2^{32}	3646716
		AC-mul + AC-inv	215282	138396002	65956	50	115702	1.78×2^{32}	3297800